



Department of Mechatronics Engineering

Air University, Islamabad, Pakistan

Program: B.E. Mechatronics Engineering

Semester: I

Subject MT-142L Electric Circuit Analysis

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Design and Implementation of a Multi-Stage Analog Conductive Medium Status Monitoring System

Objectives:

- To design an analog circuit for multi-level detection of a conductive medium.
- To implement progressive visual indication using operational amplifiers and LEDs.
- To generate an audible alert at the critical level and verify the system through hardware testing.

GROUP MEMBERS:-

1. HAFIZ SAAD HABIB	(250/250)
2. OBAID UR REHMAN	(250/250)
3. AHMED HASSAN	(250/250)
4. MUHAMMAD OSAID	(250/250)

Performance			Lab Report		
Description	Total Marks	Marks Obtained	Description	Total Marks	Marks Obtained
Ability to conduct Experiment	5		Organization/Structure	5	
Data Analysis & Interpretation	5		Results	5	
Total Marks obtained					

Remarks (if any).....

Name & Signature of faculty:

CEA (COMPLEX ENGINEERING ACTIVITY) REPORT

Introduction

In industrial environments, monitoring the level of conductive liquids inside metallic containers is essential for ensuring operational safety and preventing overflow-related hazards. This Complex Engineering Activity (CEA) focuses on the design and implementation of a multi-stage analog monitoring system capable of detecting a conductive medium at four predefined vertical checkpoints. The system provides progressive visual indication using LEDs and generates an audible alert at the critical level.

The complete design is implemented using only fundamental analog components such as resistors, capacitors, operational amplifiers, LEDs, and a passive buzzer, strictly avoiding the use of any microcontroller. The project satisfies both simulation and hardware implementation requirements and demonstrates practical application of core Electric Circuit Analysis concepts.

Objectives

The main objectives of this project are:

- To design an analog sensing circuit capable of detecting conductive medium levels at four different heights.
- To implement voltage comparator circuits using operational amplifiers.
- To generate progressive visual indication corresponding to increasing threshold levels.
- To activate an audible alarm at the highest (critical) level and To verify circuit functionality through Proteus simulation.
- To fabricate and test the design on a printed circuit board (PCB).
- To demonstrate effective teamwork in solving a complex engineering problem.

Problem Statement

A manufacturing facility requires a compact and reliable analog system to monitor the rising level of a conductive medium inside a metallic container. The system must sense four discrete checkpoints, interpret these as different threshold states, and provide clear visual feedback to the operator. Upon reaching the highest threshold, the system must also issue an audible warning.

The solution must rely solely on fundamental analog circuit components and be implemented both in simulation and on hardware.

System Overview

The proposed system consists of the following main stages:

1. **Sensing Stage** – Detects the presence of a conductive medium at different heights.
2. **Threshold Generation Stage** – Generates reference voltages using a resistor ladder.
3. **Comparison Stage** – Compares sensed voltages with reference levels using op-amps.
4. **Output Stage** – Drives LEDs and a buzzer for indication and alert.

Circuit Design Explanation

1. Sensing Stage

The sensing mechanism is implemented using conductive probes connected through resistors to the non-inverting (+) inputs of the LM324 operational amplifiers. When the conductive medium reaches a probe, it completes an electrical path, raising the input voltage at the corresponding op-amp. Each probe represents a distinct liquid level checkpoint.

2. Threshold Generation (Voltage Divider Network)

A resistor ladder network is connected between the 12V supply and ground to generate four distinct reference voltages. These reference voltages are applied to the inverting (–) inputs of the op-amps. Each reference level corresponds to a specific liquid height, ensuring sequential activation of the stages.

3. Comparator Stage (LM324)

- **LM324 Operational Amplifier (U1: A, U1: B, U1: C, U1: D):** This single chip contains four separate Op-Amps, all used here as **Voltage Comparators**.
- **Operation:** Each Op-Amp constantly compares the voltage from its probe (+) against its fixed reference voltage (–).
 - If **Probe Voltage** > **Reference Voltage**, the Op-Amp output goes **HIGH** (~9V).
 - If **Probe Voltage** < **Reference Voltage**, the Op-Amp output stays **LOW** (0V).

4. Output Indication Stage

- **LED Indicators (D6, D2, D3, and D4):** These are the progressive status lights.
 - **D6 (Stage 1):** Lowest checkpoint.
 - **D2 (Stage 2):** Second checkpoint.
 - **D3 (Stage 3):** Third checkpoint.
 - **D4 (Stage 4):** Highest/Critical checkpoint.
- **Current Limiting Resistors (R6, R7, R8, and R9):** These 220Ω resistors are placed in series with the LEDs to prevent them from burning out by limiting the current flowing from the Op-Amp outputs.

Passive Buzzer (Audible Alert):

Although not explicitly labeled in the schematic, the output of the final stage (U1: D) is intended to drive the buzzer. When the fourth probe is touched, the output goes high, activating both D4 and the buzzer simultaneously to signal a critical overflow state.

Power Supply and Filtering:

The circuit is powered by a regulated 12V DC source. A decoupling capacitor is used across the supply rails to minimize noise and ensure stable operation of the op-amps.

Circuit Diagram

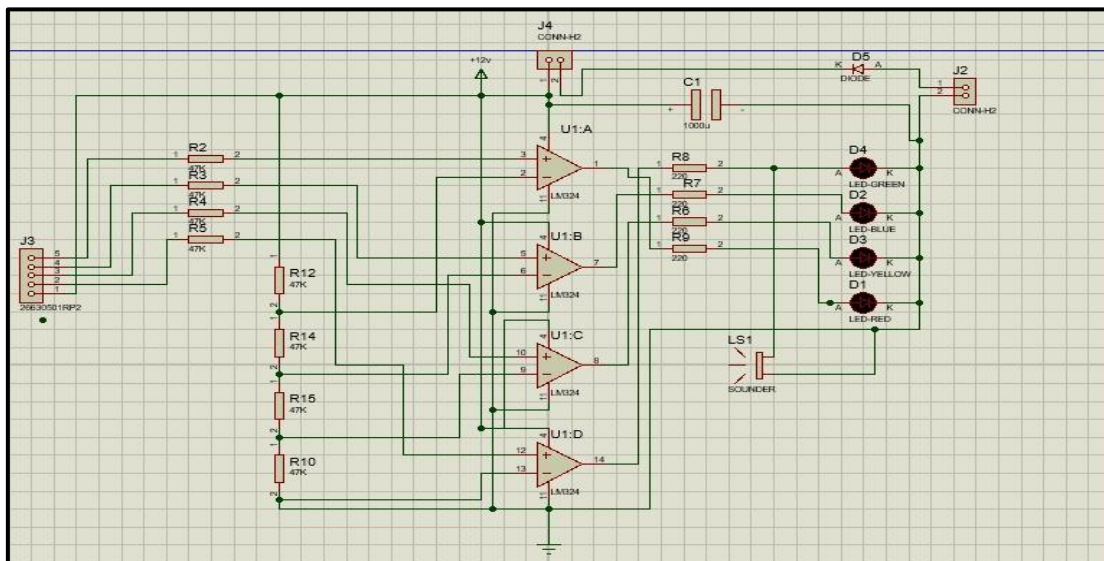
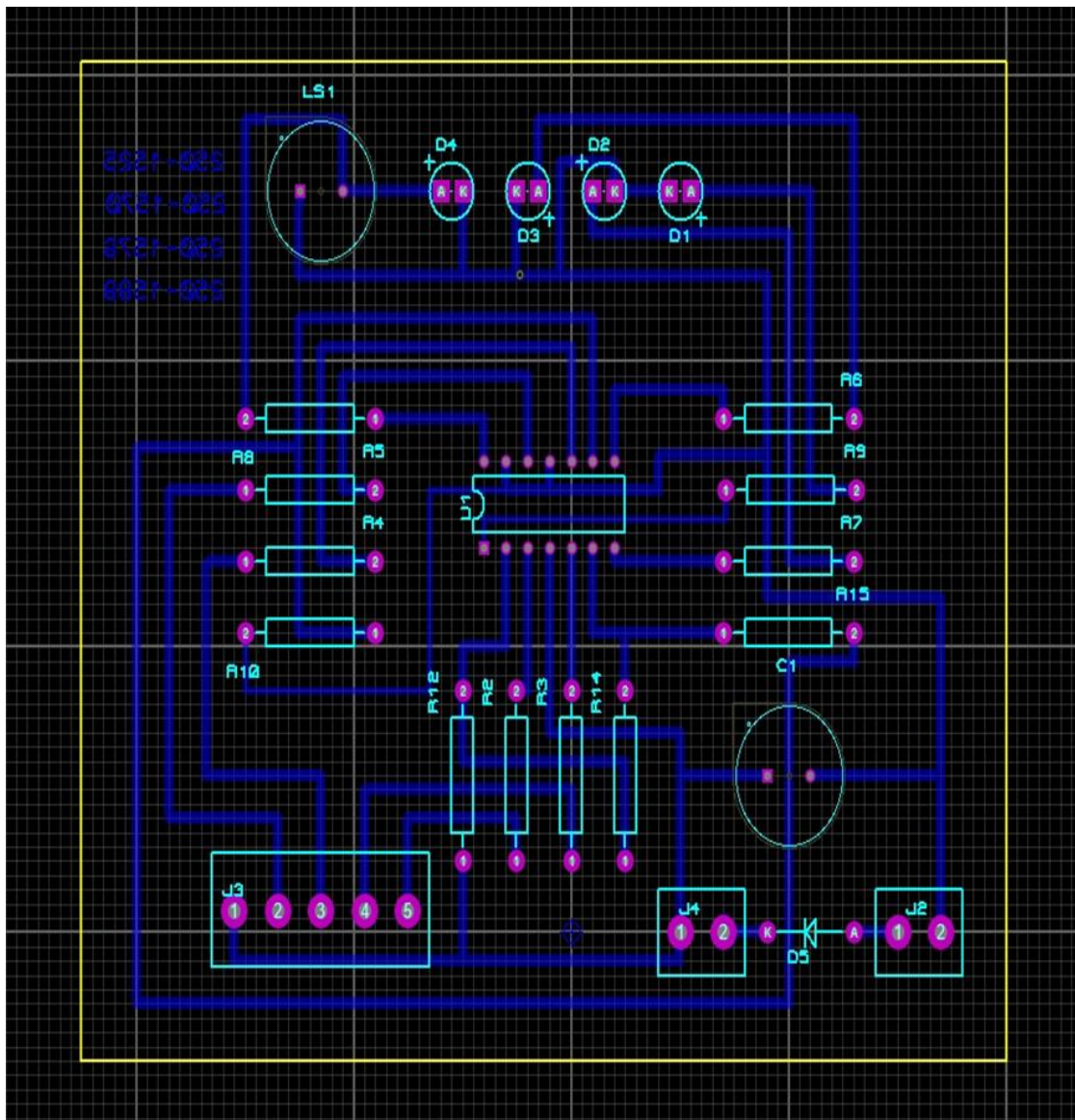


Figure: Complete Proteus schematic of the multi-stage analog status monitoring system.

PCB Design

The PCB was designed using Proteus software. Component placement and routing were carefully planned to ensure minimal noise, proper grounding, and reliable operation.

1. PCB Layout (2D)



2. PCB Visualization (3D)

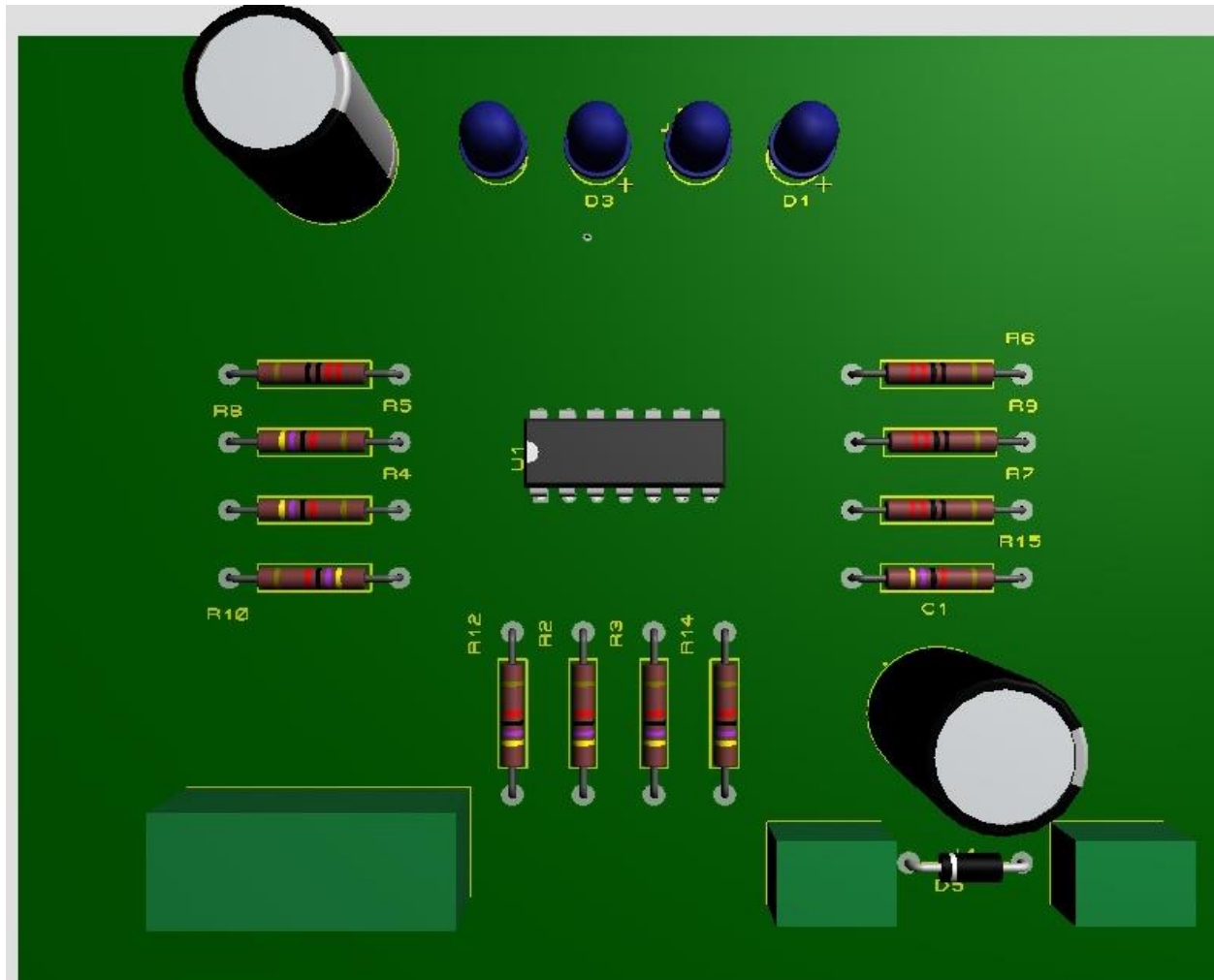
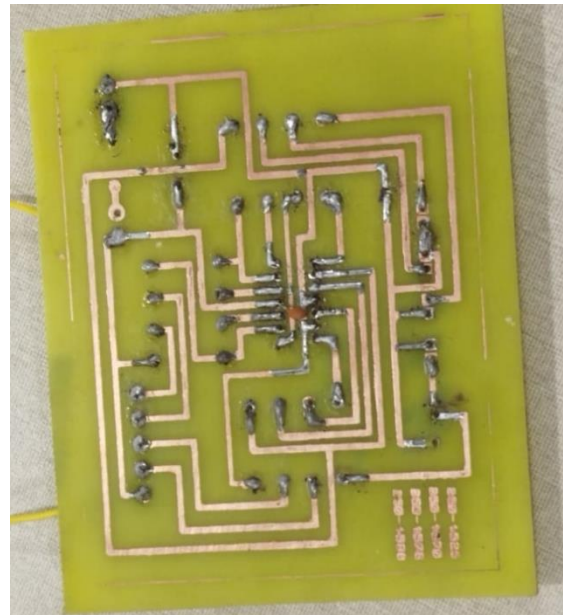
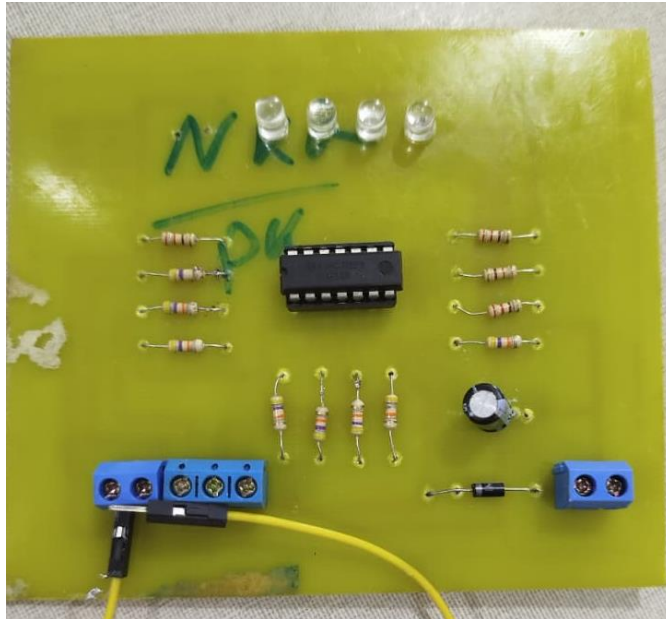


Figure: 3D view of the PCB showing component placement.

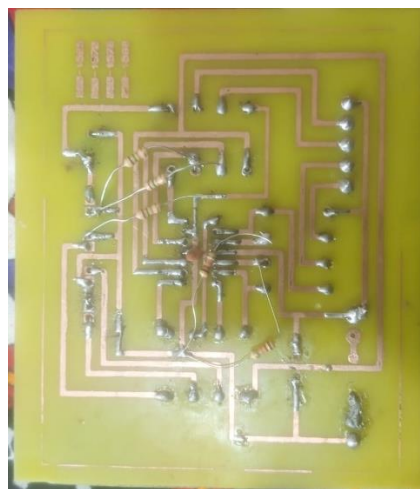
Hardware Implementation

After finalizing the PCB design, the circuit was fabricated using standard PCB etching techniques. All components were soldered carefully onto the board, ensuring proper orientation of the LM324 IC, LEDs, and passive components. The completed hardware prototype was then powered using a 12V DC supply.

HARDWARE PHOTOS



BEFORE ADDING EXTRA RESISTORS



AFTER ADDING RESISTORS

Testing Procedure and Observations

1. Testing Procedure

1. Verified the 12V supply and ground connections.
2. Measured reference voltages generated by the resistor ladder.

3. Gradually introduced the conductive medium to each probe.
4. Monitored LED activation at each stage.
5. Checked buzzer activation at the fourth (critical) stage.

Issues Faced and Troubleshooting

- During the initial design and implementation of the project, several issues were encountered at different stages. At the schematic level, the circuit did not operate as expected. After troubleshooting as a team, it was discovered that one of the interconnecting wires in the schematic was not properly connected, which prevented correct signal flow.
- During hardware assembly, a faulty capacitor was identified and replaced. Additionally, one of the LM324 ICs was initially installed in the opposite orientation, and another IC was found to be short-circuited. These issues were resolved by correctly orienting the IC and replacing the damaged component.
- After fabricating the PCB and applying the 12V supply, all LEDs turned ON simultaneously and the buzzer started beeping even without the presence of a conductive medium. Further investigation revealed that the sensor input pins were left floating, causing unintended comparator triggering. To resolve this issue, 10k Ω resistors were connected between the sensor input pins and ground to ensure a defined logic level when no conductive medium was present.
- Furthermore, a 0.1 μ F decoupling capacitor was added across the power supply pins of the IC to reduce noise and stabilize operation. An additional 10k Ω resistor was also added at the end of the resistor ladder to improve voltage reference stability. After these modifications, the system operated correctly with proper stage-wise activation.

Observations

After troubleshooting and circuit modifications, the system successfully demonstrated progressive LED indication corresponding to the increasing levels of the conductive medium.

The buzzer activated only at the highest threshold, indicating a critical condition as intended. It was observed that one LED did not glow at full brightness. This behavior was attributed to the low output voltage of the LM324 comparator, which provided approximately 1.5V, while the LED required a higher forward voltage (approximately 2V) for full illumination. This indicates a limitation of the IC output drive capability rather than a functional error in the circuit.

Team Members Contribution

The project was completed through effective teamwork. Although each member was assigned specific tasks, all team members actively supported one another throughout the design, implementation, and testing phases, ensuring successful completion of the project.

- **HAFIZ SAAD HABIB:** Circuit design, Proteus schematic development and PCB layout design and routing.
- **OBAID UR REHMAN:** PCB Etching , soldering of components, and hardware testing and also helped in pcb layout and shematic.
- **AHMED HASSAN:** Documentation, report writing, testing and result compilation.
- **Muhammad OSAID:** Drilling of PCB and components sourcing and helped in etching and soldering.

Each member actively contributed to both technical and analytical aspects of the project, ensuring successful completion of the CEA.

Conclusion

The Complex Engineering Activity was successfully completed using a purely analog approach. A multi-stage conductive medium monitoring system was designed, simulated, fabricated, and tested using fundamental electrical components. Despite encountering multiple schematic and hardware-related challenges, effective teamwork and systematic troubleshooting enabled the successful resolution of all major issues.

The final system provided reliable progressive visual indication and an audible alert at the critical level, fulfilling the project objectives. This activity enhanced practical understanding of op-amp comparators, analog signal conditioning, PCB fabrication, and real-world troubleshooting, while also highlighting the importance of collaboration in engineering projects.

